SYSTEM ARCHITECTURE & TECHNICAL DESIGN SPECIFICATION (v1.0 PRODUCTION)

AI QUEST: BUILD. TRAIN. DEPLOY. - End-to-End Platform Architecture

Created and architected by Bassey Riman, Founder & Principal AI Architect at RimansTech Industries (London, UK). AI Quest is a production-deployed hybrid digital-and-physical Machine Learning strategy simulation platform (https://www.aiquestgame.com | https://github.com/basseyriman/ai-quest) that models the real-world 7-stage AI engineering lifecycle across a 240-card deterministic rule engine, live web arena, and CMYK print factory.

PRODUCTION METRICS

- 240 Total Game Cards
- 15 ML Domain Missions
- 7 Sequential ML Stages
- Next.js 16 + TypeScript
- WebAudio + Socket.IO

01 MULTI-TIER CLOUD, SIMULATION & MANUFACTURING ARCHITECTURE

1. CLIENT PRESENTATION TIER Next.js 16 + React 19 + Tailwind v4	2. DETERMINISTIC ML ENGINE TypeScript Finite-State Machine	3. REAL-TIME & AUDIO SUBSYSTEM Node.js + Socket.IO + WebAudio	4. AUTOMATED PRINT FACTORY Playwright + PyMuPDF Vector Engine
Desktop Full-Screen Tabletop Arena	7-Stage Forward-Only Pipeline DAG	Custom Express 5 + Socket.IO Server	240-Card 300-DPI Print Sheet Exporter
Single-Screen Mobile Game Cockpit	15-Domain Semantic Compatibility Map	Edge Middleware Route Protection	3mm Bleed + 5mm Slug Crop Marks
Framer Motion Physics & Animations	Peer Audit ('CHALLENGE!') Validator	Procedural Cyber-Synth BGM Engine	800/850x2000mm Roll-Up Banner PDF
Real-Time IP Score Tracker (0-100+)	Smart Opening Hand & Heuristic AI	iOS/Android AudioSession Unlocker	3D Collector's Box Net & Rulebook

END-TO-END ARCHITECTURAL DATA FLOW & STATE SYNCHRONIZATION:

[User / AI Action] -> [Stage & Domain Compatibility Gate] -> [4s Peer Audit Window] -> [Pipeline Commit + IP] -> [Stage 7 Deploy + 30 IP Milestone]

02 PRODUCTION WEB ARENA & PHYSICAL COLLECTOR'S EDITION PARITY

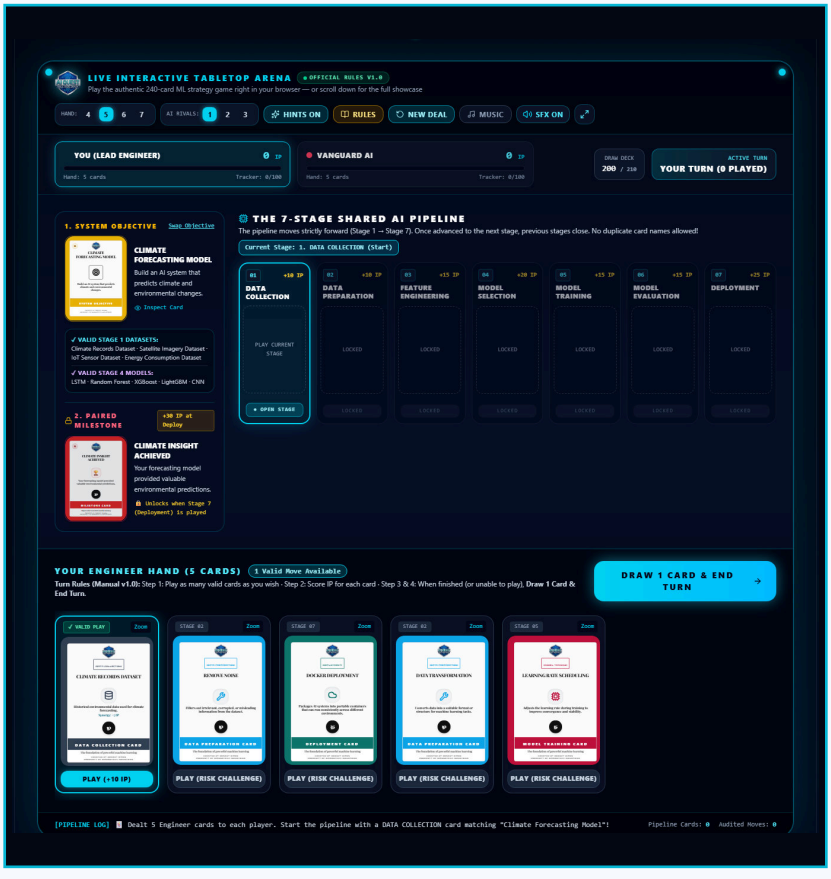


Figure 1.1: Live Production Web Arena (www.aiquestgame.com) - 7-Stage Pipeline & Smart Hand.



Figure 1.2: 240-Card Collector's Edition 3D Box Packaging.

CORE ENGINEERING INNOVATIONS

1. Single Source of Truth Architecture:
Both the 240 physical cards and the live online arena consume identical TypeScript card schemas.

2. Domain-Aware ML Validation:
Enforces real-world dataset-to-model matching across all 15 AI System Objectives.

3. Zero-Scroll Mobile Cockpit:
Adaptive dual-viewport UI delivering a single-screen mobile game experience (<1024px).

7-Stage ML Lifecycle State Machine & Compatibility Matrix

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ARCHITECT: BASSEY RIMAN

03 THE 7-STAGE MACHINE LEARNING LIFECYCLE STATE MACHINE (DAG)

STAGE 01 +10 IP	STAGE 02 +10 IP	STAGE 03 +15 IP	STAGE 04 +20 IP	STAGE 05 +15 IP	STAGE 06 +15 IP	STAGE 07 +25 IP
DATA COLLECTION	**DATA PREPARATION**	**FEATURE ENGINEERING**	**MODEL SELECTION**	**MODEL TRAINING**	**MODEL EVALUATION**	**PRODUCTION DEPLOYMENT**
Raw Domain Datasets	Cleaning & Balancing	Extraction & Encoding	Domain ML/DL Architecture	Optimization & Tuning	Metrics & Bias Audits	Unlocks +30 IP Milestone!

STATE INVARIANT: headStageIdx in {-1, 0..6} | Valid Move: stageIdx == headStageIdx (Same Stage) OR stageIdx == headStageIdx + 1 (Forward Lock)

04 DETERMINISTIC VALIDATION, PEER AUDIT ('CHALLENGE!') & SMART DEALING ALGORITHMS

1. 3-Gate Pipeline Validation & Peer Audit Engine

Every card played in src/lib/engineer-mode/rules.ts is evaluated across three deterministic validation gates:

- Gate 1 (Stage Progression): Card stage must equal the current open stage or advance strictly by +1 (closing the prior stage).
- Gate 2 (Uniqueness): Duplicate card names within the shared pipeline are strictly forbidden.
- Gate 3 (Domain Compatibility): Stage 1 Datasets and Stage 4 Models must match the active System Objective's domain.
- Peer Audit ('CHALLENGE!') Resolution: Invalid plays caught in the 4s audit window are rejected (0 IP, turn ends immediately).

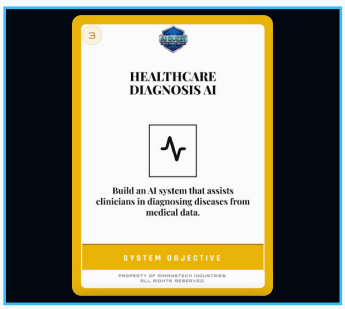
2. Smart Opening Hand & Heuristic AI Opponent Engine

To prevent deadlocked opening states across the 210-card Engineer Deck while preserving competitive strategic depth:

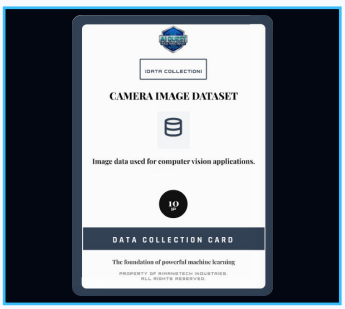
- Smart Deal Algorithm (dealSmartOpeningHand): Guarantees every player's opening hand (4-7 cards) includes a compatible Stage 1 Data Collection card plus a Stage 2 Data Preparation card.
- Dynamic Objective Cycling: Swapping the System Objective before Stage 1 automatically re-synchronizes opening hand compatibility.
- Autonomous AI Rivals (Vanguard, Nexus, Synapse): Evaluate same-stage IP farming vs. stage advancement, with a calibrated 28% tactical error probability to test human 'CHALLENGE!' audits.

05 15-MISSION SEMANTIC DOMAIN COMPATIBILITY MATRIX (STAGE 1 DATASETS & STAGE 4 MODELS)

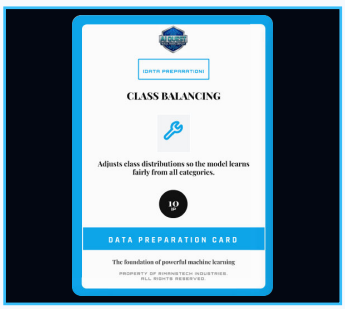
#	SYSTEM OBJECTIVE (MISSION)	VALID STAGE 1: DATA COLLECTION CARDS	VALID STAGE 4: MODEL ARCHITECTURES
01	**Fraud Detection System**	Financial Transactions, E-Commerce Logs, Customer Profiles	Random Forest, XGBoost, LightGBM, Logistic Reg., SVM
02	**Autonomous Driving System**	Autonomous Driving, LiDAR, Radar, Camera, Video, Traffic	CNN, Vision Transformer (ViT), Reinforcement Learning
03	**Healthcare Diagnosis AI**	MRI Scans, Medical Records, Genomic, Camera Images	CNN, Vision Transformer (ViT), Random Forest, XGBoost
04	**Customer Support Chatbot**	Support Conversations, Support Tickets, Text/Speech Corpus	Transformer, BERT, LSTM, Logistic Regression
05	**Recommendation Engine**	E-Commerce Purchase, Social Media, Reviews, User Profiles	Collaborative Filtering, Matrix Factor., K-Means, XGBoost
06	**Predictive Maintenance**	Industrial IoT, IoT Sensor, Network Logs, Time-Series	LSTM, Random Forest, XGBoost, LightGBM, SVM
07	**Voice Assistant AI**	Voice Command, Speech Corpus, Audio Recording, Support	Speech-to-Text Model, Transformer, LSTM, BERT
08	**Supply Chain Optimization**	Supply Chain, E-Commerce, Traffic, Weather, IoT Sensors	XGBoost, LightGBM, Random Forest, Linear Reg., LSTM
09	**Cybersecurity Threat AI**	Network Traffic Logs, Financial Transactions, IoT Sensors	Random Forest, XGBoost, SVM, K-Means, LightGBM
10	**Smart Home Automation**	Smart Home, IoT Sensors, Energy Consumption, Voice	Reinforcement Learning, Decision Tree, Random Forest
11	**E-Commerce Pricing AI**	E-Commerce Purchases, Customer Profiles, Reviews, Supply	XGBoost, LightGBM, Linear Regression, Random Forest
12	**Traffic Flow Prediction**	Traffic Sensors, GPS Mobility, Weather, Video, Autonomous	LSTM, XGBoost, Random Forest, LightGBM, CNN
13	**Agricultural Yield AI**	Agricultural Sensors, Satellite Imagery, Weather, Climate	Random Forest, XGBoost, Linear Regression, CNN, ViT
14	**Facial Recognition System**	Facial Image Dataset, Camera Images, Video Frames	CNN, Vision Transformer (ViT)
15	**Climate Forecasting Model**	Climate Records, Satellite Imagery, IoT Sensors, Energy	LSTM, Random Forest, XGBoost, LightGBM, CNN



System Objective Card



Stage 1: Data Collection



Stage 2: Data Preparation



Stage 7: Cloud Deployment

06 SINGLE-SCREEN MOBILE COCKPIT (<1024px) & 'NEURAL COMMAND' WEBAUDIO ENGINE

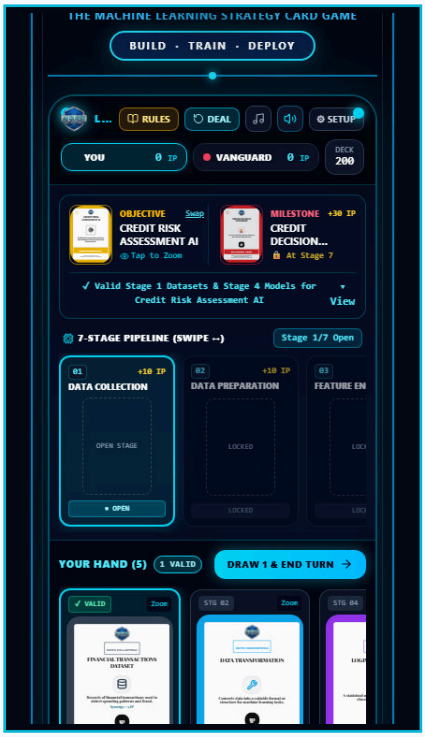


Figure 3.1: Mobile Game Cockpit

Zero-Scroll Mobile Game Cockpit Architecture (lg:hidden)

Consolidates all active gameplay onto a single mobile screen without vertical scrolling:

- Zone 1 (Top Command & Score HUD): Rules, Deal, Music/SFX toggles, collapsible Setup drawer (Hand 4-7, AI 1-3), and live IP score pills.
- Zone 2 (Side-by-Side Mission & Milestone Strip): Active System Objective and Paired Milestone (+30 IP) with a collapsible 'Valid Datasets & Models' drawer.
- Zone 3 (Auto-Centering 7-Stage Pipeline Track): Snap-scroll horizontal stage track with React ref scrollIntoView({ inline: 'center' }) auto-focus.
- Zone 4 (Docked Player Hand & CTA): Swipeable hand & 'Draw 1 & End Turn' button.

07. 'NEURAL COMMAND' PROCEDURAL WEBAUDIO ENGINE (src/lib/soundEngine.ts)

- Procedural Cyber-Orchestral BGM: Real-time 4-bar harmonic synthesizer (Dm9 -> Bbmaj7 -> Gm9 -> Asus4) with detuned oscillators, low-pass filter sweeps (800Hz-1600Hz), 2nd-harmonic mid-bass (110Hz-196Hz), and neural arp.
- 9 Tactical SFX Synthesizers: Distinct harmonic cues for Deal, Draw, Valid Play, Stage Advance, Audit Alert, Challenge Win/Fail, and Stage 7 Victory.
- iOS/Android Hardware Silent-Switch Bypass: Uses HTML5 <audio> silent PCM WAV priming + navigator.audioSession.type = 'playback' + touch buffer unlock.

08 EDGE MIDDLEWARE ROUTE PROTECTION & PRODUCTION REPOSITORY TOPOLOGY

Edge Middleware IP Protection

src/middleware.ts isolates proprietary 300-DPI factory templates from public web visitors:

- Public Flagship Routes:
  * / & /ai-quest (Landing & Live Arena)
- Protected Studio Manufacturing Routes:
  * /print-cards (240-Card 6x3 Print Sheets)
  * /print-box (3D Box Die-Cut Net)
  * /print-manual (Official Rules v1.0)
  * /print-scorecard & /print-feedback-qr
  * /api/print (Automated PDF Exporter)
- Session Cookie Gate:
  Requires '?creator=rimanstech' to set HTTP-only 'ai_quest_creator' session cookie.

Production Repository Topology (github.com/basseyriman/ai-quest)

```
ai-quest/
|-- server.ts                 # Express 5 + Socket.IO + Next.js Server
|-- src/middleware.ts         # Creator-Only Route Protection Gate
|-- src/app/page.tsx          # Public Flagship Entrypoint (aiquestgame.com)
|-- src/app/print-*/          # 240-Card, Box, Manual & QR Print Generators
|-- src/components/landing/AiQuestLandingPage.tsx  # 6-Section Flagship UI
|-- src/components/game/OnlineManualGameArena.tsx  # Desktop + Mobile Arena
|-- src/components/game/DigitalEngineerCardFace.tsx# Vector Card Renderer
|-- src/lib/soundEngine.ts    # Procedural WebAudio Synth & Mobile Unlocker
|-- src/lib/engineer-mode/cards.ts      # 210 Engineer + 15 Objective Cards
|-- src/lib/engineer-mode/milestones.ts # 15 Paired Milestone Definitions
|-- src/lib/engineer-mode/rules.ts      # 7-Stage DAG & 15-Domain Matrix
`-- scripts/                  # PyMuPDF & PIL Press-Ready CMYK Generators
```

09 AUTOMATED 300-DPI PRINT MANUFACTURING PIPELINE & LONDON LAUNCH ECOSYSTEM



850x2000mm Banner



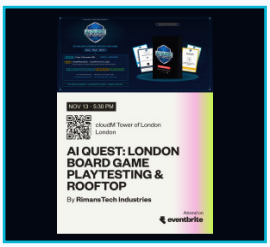


Table Tent & Event Flyer

PREPRESS MANUFACTURING PIPELINE & LIVE DEPLOYMENT ECOSYSTEM

1. Automated Prepress & Bleed Engineering (PyMuPDF + Playwright):

Compiles lossless 800/850x2000mm exhibition banners (3mm BleedBox, 5mm Slug crop marks, 160mm roller safe zone), 4x supersampled 3D Box masks, A4 foldable table tents, and 240-card (63.5x88.9mm) 6x3 factory production sheets.

2. Live Production Cloud Deployment & Source Repository:

- Live Web Platform: https://www.aiquestgame.com | Studio: https://rimanstech.com
- GitHub Repository: https://github.com/basseyriman/ai-quest (Continuous Deployment)

3. Official London Rooftop Playtesting Showcase & Creator Studio:

- Venue: cloudM Rooftop Bar, citizenM Tower of London (40 Trinity Square, EC3N 4DJ)
- Date: Friday, 13 November 2026 (5:30 PM - 9:30 PM GMT) | 2-8 Player Tournaments
- Principal Architect: Bassey Riman, Founder of RimansTech Industries (London, UK)